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All Reviews of cec_pap419

details

Decision: **Reject**

Submission: Optimizing Surgical Patient Flow through Operating Room Scheduling: A Use Case Study

Contributors: Abdelaidoum, Boubchir, Daachi, Dagneaux, Madani, de Coüasnon

Key for the below column headings show

Summary of Reviews of cec_pap419s1

Reviewer	rec	conf	oral	bp	indust
Reviewer 1	accept (4)	confident (3)	oral (1)	No (0)	No (0)
Reviewer 2	weak reject (2)	somewhat confident (2)	poster (0)	No (0)	Yes (1)
Reviewer 3	weak accept (3)	somewhat confident (2)	poster (0)	No (0)	Yes (1)
Reviewer 4	weak reject (2)	confident (3)	poster (0)	No (0)	No (0)
Averages:	**2.8**	**2.5**	**0.2**	**0.0**	**0.5**

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Committee Comments · jump down

Full Reviews

Reviewer 1

top

Novelty and Significance

monospace

The work of (Ahmed & Ali, 2020), “Modeling patient preference in an operating room scheduling problem,” should be cited in this paper, as it appears to be the first study to explicitly incorporate “patient preferences” into OR scheduling (referred to as “patient choice” by the authors of this paper). In (Ahmed & Ali, 2020), patient preference refers to the selection of surgeons based on personal criteria. In contrast, in the present paper, patient preferences/choices are modeled by offering each patient two or three distinct time slots from which to choose. Although the interpretations differ, both approaches address the integration of patient considerations into OR scheduling, and the earlier contribution should therefore be acknowledged.

I mention this point because the novelty of this paper lies in the idea that previous studies did not consider patients choosing their preferred time slots when solving the MILP. This appears to be correct but should be carefully verified.

The work by (Xue Yang and Haichao Liu and Yang Wang and Abraham P. Punnen and Haibo Wang) 2025: “Integrating Patient Satisfaction into Operating Room Scheduling Decisions”, should also be taken into consideration.

Relevance

monospace

I find this work relevant to CEC, particularly due to the simplicity of the Flexibility-First Heuristic for generating assignments on the fly.

Rigour and Soundness

markdown monospace

- The experiments are generally easy to follow and could be readily understood by a new reader. However, there is some redundancy, for instance between Fig. 2 and Table 3. It is not clear what additional insight the histogram provides.

- There appear to be errors in Fig. 4 regarding the parameterization (2, 20, 3), which is shown with a runtime of 600s. It seems this should instead be (2, 40, 3), positioned toward the far right.

- The last paragraph before the conclusion, “The computational performance aligns ...” seems largely redundant with previous observations and analysis. Consider either revising it to provide a deeper insight or removing it completely.

Structure and Readability

markdown monospace

Overall, the paper reads very well. I have a few suggestions for improving readability:

- You may consider including the term "patient choice" in the paper title to better reflect the main contribution of the work.
- The Related Work section currently reads like one long, hard-to-follow paragraph. It would be clearer if broken into medium-length paragraphs, each centered on a specific research topic with a clear purpose for inclusion.
- In Section C.1 (Objective): Try to briefly describe each equation (1)–(5), as you have already done for the constraints. This greatly helps in explaining the MILP.
- For equations (10) and (11): I would suggest reversing the definitions of EA_{p,i} and LD_{p,i}, for eg. : EA_{p,i} will equal 1 when reprogramming is allowed.
- For Table 4, adding a vertical line would clearly separate the parameter columns (Physicians, Patients, Blocks) from the result columns (Events, Patients With Events, ...).

Invite to TEVC

monospace

no

If so, why and what extensions would you recommend?

Review to TEVC

Other Comments for Authors

None

Scored Review Questions

	rec	conf	oral	bp	indust
cec_pap419s1	accept (4)	confident (3)	oral (1)	No (0)	No (0)

Reviewer 2

top

Novelty and Significance

monospace

The paper addresses a meaningful and practically relevant problem: integrating patient preference into OR scheduling, an aspect often neglected in operations research literature. The patient-centered framing and flexibility-first heuristic represent a genuinely humanistic contribution to a field dominated by pure efficiency optimization. However, the technical novelty is modest. The MILP formulation is standard, and the iterative heuristic is intuitive but not algorithmically sophisticated. The real-world deployment at CHU Montpellier adds applied significance. The paper's value lies more in its sociotechnical integration than mathematical innovation, which may limit its appeal at a primarily computational intelligence venue.

Relevance

monospace

This paper is only marginally relevant to CEC. The core methodology is classical operations research — MILP with a greedy heuristic — with no evolutionary computation, neural networks, or computational intelligence techniques involved. The authors mention genetic algorithms briefly in future work, but the current submission does not employ them. While healthcare optimization is a valid application domain for CEC, the paper lacks the evolutionary or nature-inspired algorithmic content expected at this venue. It would be better suited to journals or conferences focused on healthcare operations management or applied OR.

Rigour and Soundness

monospace

The experimental design is reasonable for a use-case study. Real hospital data from 134 patients over three weeks grounds the results in practice, and the 20-run benchmark replication adds statistical reliability. However, there are notable weaknesses: the 600-second timeout for P=40 with 15% optimality gaps is presented as acceptable without justification of why this gap is clinically tolerable. The Week 3 utilization drop to 39% is a structural artifact of the redistribution, not a failure, but it is insufficiently explained. No comparison against alternative heuristic or metaheuristic approaches is provided, making it difficult to assess the heuristic's relative performance.

Structure and Readability

monospace

The paper is clearly written and well-organized. The workflow diagram in Figure 1 effectively communicates the iterative process. Tables I and II are thorough though dense; splitting them could improve readability. The mathematical formulation is presented cleanly. Figure 4's logarithmic runtime plot is informative. The conclusion is appropriately concise. Minor issues include a typographical error in the abstract ("surgical cate" instead of "surgical care") and occasional informal phrasing. The related work section is comprehensive and well-cited. Overall the paper reads clearly and accessibly for a broad audience.

Invite to TEVC

monospace

no

If so, why and what extensions would you recommend?

Review to TEVC

Other Comments for Authors

monospace

The work is practically valuable and the patient-centered framing is commendable. To strengthen the paper for a computational intelligence venue, the authors should consider incorporating a metaheuristic or evolutionary component and providing a direct baseline comparison. Clarifying the clinical acceptability of the 15% optimality

gap and better explaining the Week 3 utilization drop would also improve rigor. Fixing the abstract typo ("surgical cate") is recommended.

Scored Review Questions

	rec	conf	oral	bp	indust
cec_pap419s1	weak reject (2)	somewhat confident (2)	poster (0)	No (0)	Yes (1)

Reviewer 3

top

Novelty and Significance

monospace

The paper demonstrates strong novelty in the problem formulation by providing a detailed modeling of the scheduling problem in the orthopedic department of a hospital. However, the methodological novelty of the proposed approach needs to be further clarified, particularly in comparison with existing solution methods.

In addition, the experimental validation is conducted on only a single case by comparing the results with real-world data. While this provides some practical insight, more comprehensive experimental validation or further comparative studies would strengthen the credibility and reliability of the conclusions.

Relevance

monospace

The paper is highly relevant to Real-World Applications, as it models a practical scheduling problem in the orthopedic department of a hospital. It is also closely related to Heuristics and Combinatorial Optimization, since the study formulates the problem as a combinatorial optimization model and proposes a heuristic method to solve it.

Although the modeling section includes detailed penalty term formulations, the methodological part lacks an explicit constraint-handling mechanism. Therefore, its relevance to Constraint Handling is relatively limited.

Rigour and Soundness

monospace

The overall rigor and soundness of the paper are moderate. Although the problem formulation is described in considerable detail, several other aspects of the manuscript require further improvement. For instance, the Introduction section contains only a single citation, which is insufficient to adequately position the work within the existing literature.

In the experimental section, validation appears to be conducted by comparing the results with only one real-world case. There seems to be limited comparison with existing methods. In addition, the presentation of experimental results is somewhat coarse. Providing more detailed performance metrics or statistical analysis would improve the rigor and reliability of the study.

Structure and Readability

monospace

The overall structure and logical flow of the manuscript need improvement. There are numerous minor errors and several sections that appear insufficiently organized or not fully developed. As a result, the readability of the paper is relatively weak. Careful revision of the structure and thorough proofreading are strongly recommended to enhance clarity and coherence.

Invite to TEVC

monospace

no

If so, why and what extensions would you recommend?

Review to TEVC

Other Comments for Authors

markdown monospace

This study presents a patient-centered scheduling framework for the orthopedic department of a hospital and provides a detailed problem formulation. The experimental results demonstrate promising performance. Nevertheless, the novelty of the proposed algorithm needs to be further clarified. In addition, the manuscript contains several minor errors and inconsistencies that require careful correction:

1. There is only one citation in the Introduction section, which seems insufficient to adequately support the background and motivation of the study.
2. The first paragraph of the Related Work section is relatively long, which affects readability.
3. The placement of table titles is inconsistent: the titles of Tables I and II appear below the tables, while those of Tables III and IV appear above. Please unify the formatting throughout the manuscript.
4. In Section III.D, the novelty of the proposed algorithm is not sufficiently clarified. In particular, it would be helpful to explain how the method improves constraint handling or contributes to the treatment of the penalty terms.
5. The decision variable is defined as X, whereas Algorithm 1 uses A, and the variables h, d, and b are not mentioned in the algorithm description. This inconsistency may cause confusion. It would be helpful to explicitly explain the relationship between A and X, and clarify the roles of the omitted variables.
6. It is unclear whether the input S_p of the algorithm is obtained from solving the MILP model using a solver or is manually predefined. Please clarify its source. If it is manually specified, the experimental section should clearly describe how S_p is generated and configured.
7. In the experimental section, what is the difference between the conclusions drawn from Table I and Fig. 2? If they convey the same results, please justify the necessity of presenting both.

8.There are some minor issues and possible errors:
In Table I, the notation “P in D^part” uses the same letter P as the “patient count P” in Section IV.C, which may cause confusion.
In Section III.D, “ppp” should be corrected to “p”.
In Section III.D, Algorithm 1: do you mean “as shown in Algorithm 1”?
In Section IV.B, “3%” should likely be “39%”.
In Fig. 4, some descriptions do not correspond correctly to the related table. Please verify consistency.

9. There are some writing and organization Issues:
   In Section IV.C, the final paragraph largely repeats conclusions already stated in the previous two paragraphs. It may benefit from condensation or reorganization.
   In the Conclusion section, the last two to three paragraphs appear insufficiently structured and could be further refined for logical coherence.

Additionally, it is recommended that the authors carefully proofread the manuscript to identify and correct any other potential inconsistencies or unfinished content.

Scored Review Questions

	rec	conf	oral	bp	indust
cec_pap419s1	weak accept (3)	somewhat confident (2)	poster (0)	No (0)	Yes (1)

Reviewer 4

top

Novelty and Significance

monospace

This paper presents a use case study for a patient-centered scheduling framework developed for an operating room in a specific hospital. The initial objective is to propose an algorithm that optimizes operational excellence and patient care for elective orthopedic surgery. The algorithm consists of three steps: input of patient and physician data, optimization of allocation through a mixed integer linear program (MILP) model that handles resource constraints and patient availabilities, and an iterative flexibility first heuristic to select and confirm patient selections using a high availability score for each patient.

The novelty lies in considering patient preferences and how they improve the management of healthcare operations, which is important and interesting. However, technically it does not have a significant contribution in terms of models and algorithms.

Relevance

monospace

The practical relevance is unquestionable and confirmed by the results obtained when compared to a real-world case previously applied. It is relevant for better patient care in a social context, but not for what is expected at a conference like the CEC. The MILP model is well-known in the literature, and the heuristic is simple, intuitive, and does not offer a significant contribution in terms of structure and formalism.

Rigour and Soundness

monospace

The mathematical formulation is well known in the literature of classic timetabling problems involving resource utilization and balancing preferences, avoiding clashes. The proposed heuristic does not follow any formal structure, makes use of a controversial criterion for selecting and ordering patients, and a procedure for penalizing and prioritizing objectives. The experimental section is the weakest part. The instances tested are small, it would be

necessary to explain how the solution in the real case was obtained, and the comparisons are weak. The study lacks a state-of-the-art comparison even with techniques that do not consider patient participation in decision-making.

Structure and Readability

monospace

The paper is easy to read, clear and objective, but it lacks a greater technical and scientific contribution.

Invite to TEVC

monospace

no

If so, why and what extensions would you recommend?

Review to TEVC

Other Comments for Authors

None

Scored Review Questions

	rec	conf	oral	bp	indust
cec_pap419s1	weak reject (2)	confident (3)	poster (0)	No (0)	No (0)

Committee Comments

top

Committee Comments for Authors

None

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